

Lecture - 2

❖ Evolution of Programming Languages

Programming languages have evolved from **machine-oriented languages** to **human-friendly and problem-oriented languages**.

➤ First Generation Language (1GL) – Machine Language

- Machine language is the lowest-level programming language.
- Instructions are written using 0s and 1s (binary).
- It can be directly understood and executed by the CPU.
- It is very fast but extremely difficult for humans to write, understand, and debug.
- It is machine dependent.

Example:

10110000 01100001

➤ Second Generation Language (2GL) – Assembly Language

- Assembly language uses short words called mnemonics instead of binary.
- Examples of mnemonics: MOV, ADD, SUB, JMP.
- It is easier to understand than machine language.
- An assembler converts assembly language into machine code.
- It is generally hardware/machine dependent.

Example:

MOV A, B

ADD A, C

➤ Third Generation Language (3GL) – High-Level Language

- High-level languages are designed to be easy for humans to read, write, and understand.
- They use English-like statements and mathematical expressions.
- They are generally more portable than assembly language.
- A compiler or interpreter translates the program into a form the computer can execute.

Examples: C, C++, Java, Python, FORTRAN, Pascal.

Example in C:

sum = a + b;

➤ Fourth Generation Language (4GL)

- 4GL provides a higher level of abstraction than 3GL.
- It requires fewer lines of code to perform tasks.
- It is often used for database operations, report generation, and data analysis.
- The programmer specifies what is required, rather than describing every step to achieve it.
- SQL is a common example.

Example:

SELECT name, marks FROM student WHERE marks > 80;

➤ Fifth Generation Language (5GL)

- 5GL is mainly associated with Artificial Intelligence (AI) and logic/constraint-based programming.

- The programmer specifies the problem, rules, constraints, or desired result.
- The system determines how to arrive at the solution.
- Prolog is a classic example.
- It is used in areas such as AI, expert systems, and knowledge-based applications.

Example idea:

Facts + Rules + Query → Solution

Comparison Table

Generation	Language	Main Idea	Example
1GL	Machine Language	0s and 1s	10110000
2GL	Assembly Language	Mnemonics	MOV, ADD
3GL	High-Level Language	Human-readable programming	C, Java, Python
4GL	Very High-Level Language	Specify what is needed	SQL
5GL	AI/Logic Language	Specify problem, rules, constraints	Prolog

❖ Language Translators

These are the programs which are used for converting the programs in one language into Machine language instructions, so that they can be executed by the computer.

1. **Assembler:** It is a program which is used to convert the assembly level language programs into machine language.
2. **Compiler:** It is a program which is used to convert the high level language programs into machine language.
3. **Interpreter:** It is a program, it takes one statement of a high level language program, translates it into machine language instruction and then immediately executes the resulting machine language instruction and so on.